

GOKHALE MEMORIAL GIRLS' SCHOOL

PRE BOARD EXAMINATION - I (2025-26)

CLASS - X

SCIENCE

Time - 1 Hr. 30 mins.

Full Marks - 40

General Instructions :

- i. This question paper has 20 questions in 3 sections. Section A is Biology, Section B is chemistry and Section C is physics
- ii. All questions are compulsory. However internal choice is provided in some questions.

Section - A (~~Boiology~~) Biology

1. Which hormone regulates carbohydrate, protein and fat metabolism in the body?
a) Growth hormone b) ☒ Thyroxine
c) Insulin d) Adrenaline [1]
2. A boy starts developing facial hair and a deeper voice during puberty. Which hormone is responsible?
a) Estrogen b) ☒ Testosterone
c) Insulin d) Oxytocin [1]

P. T. O

3. Which method of contraception is most effective in preventing pregnancy and sexually transmitted infection. [1]

a) ☒ Condom

b) Copper - T

c) Oral pills

d) Surgical methods

4. Assertion (A) : Variations which occur during reproduction help in the survival of the species in changing environment.

Reason (R) : DNA copying during reproduction is not perfectly accurate.

a) ☒ Both 'A' and 'R' are true, and 'R' is the correct explanation of 'A'.

b) Both 'A' and 'R' are true, but 'R' is not the correct explanation of 'A'.

c) 'A' is true, but 'R' is false.

d) 'A' is false, but 'R' is true. [1]

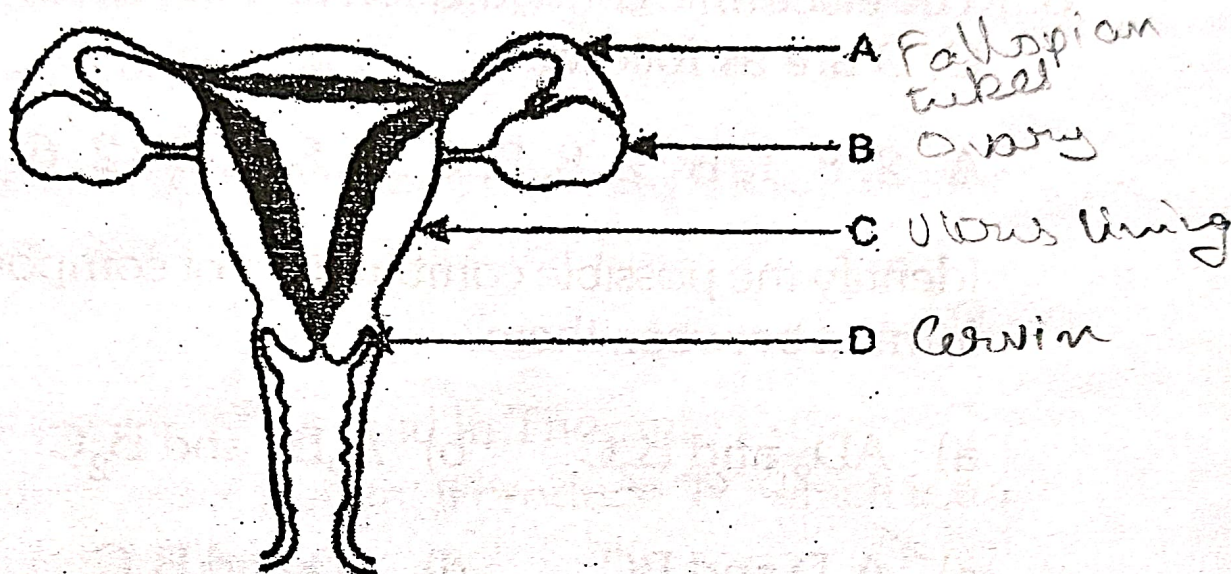
5. Yeast and Amoeba are both unicellular organisms. How do their mode of reproduction differ? [2]

6. What is feedback mechanism? Explain with an example how it regulates hormone secretion. [3]

OR

- a) How is goitre caused? *deficiency of iodine*
- b) How blood sugar is regulated in our body?
Name the disease and treatment involved with this. *diabetes* *insulin* [1+2]

7.



- a) Label the parts marked as A to D.
- b) What happens to the lining of uterus
- i) If fertilisation occurs
- ii) If no fertilisation occurs [2+3]

OR

- a) Draw a diagram to show germination of pollen on stigma. Label the following parts.

Pollen grain, stigma, male germ - cell, pollen tube, ovary, female germ - cell.

- b) Mention two points when vegetative propagation is preferred in plants over sexual reproduction. [3+2]

Section - B (Chemistry) [13 Marks]

8. The electronic configuration of 4-elements A, B, C, D are as follows:

A : 2, 8, 1, B : 2, 8, 2, C : 2, 8, 7, D : 2, 6

Identify the possible combination of compounds formed between them.

- a) AD_2 and B_2C b) A_2D_3 and B_2C
c) A_2D and BC_2 d) AD and B_2C_3 [1]

9. Which of the following pairs are not bases?

a) NH_4OH , $NaOH$ b) KOH , $Zn(OH)_2$

c) C_2H_5OH , CH_3COOH

d) $Ca(OH)_2$, CaO [1]

10. There are two statements, Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

a) Both A & R are true, and R is the correct explanation of A.

b) Both A & R are true, but R is not the correct explanation of A.

c) A is true, R is false.

~~d) A is false, R is true.~~

Assertion (A) : The next higher homologue of Ethanol is Pentanol.

Reason (R) : Any two members in a homologous series differ in the molecular mass by 14U.

[1]

11. The metal, 'M' is used in Thermite Process. On heating with Oxygen, it produce 'N' which is an amphoteric oxide.

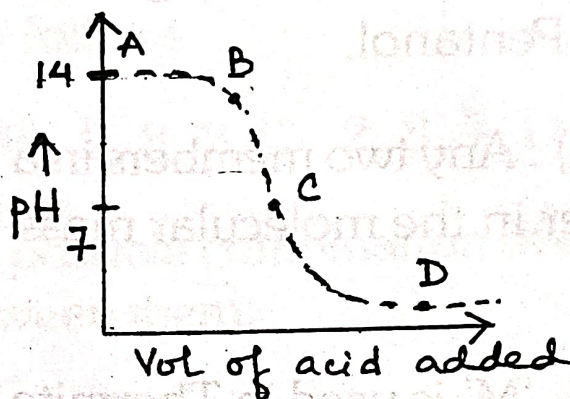
a) Identify 'M' & 'N'.

b) Write the balanced chemical reaction of 'N' with dilute Sulphuric Acid and Sodium Hydroxide solution separately. [1+1=2]

12. An ore, A of an amphoteric metal of medium reactivity on treatment with dilute hydrochloric acid, produces brisk effervescence. How is the metal extracted from the ore A? Write the steps involved, giving balanced chemical equations. [1+1=2]

13. i) There are five samples P, Q, R, S and T that change the colour of universal pH - paper to purple, reddish - pink, green, orange and blue respectively. Their pH - values were recorded as 12, 2, 7, 6 and 10.5 respectively. Which of the samples will have the maximum Hydrogen - ion concentration?

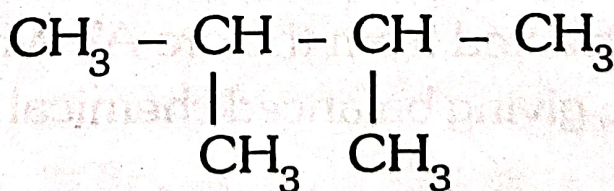
ii)



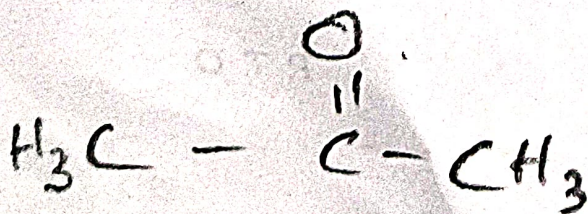
The graph given here depicts the neutralisation reaction when acid is added to an alkali. Which point on the graph denotes the presence of both acid and salt?

- iii) Why do walls immediately acquire a white colour on applying a coating of Slaked lime?
[1+1+1=3]

14. i) Draw the structural formula of Propanone.
ii) Write the IUPAC name of the following compound.



Sci. (X) — Pg. 6



- iii) Select the saturated hydrocarbon from the following list :



[1+1+1=3]

OR

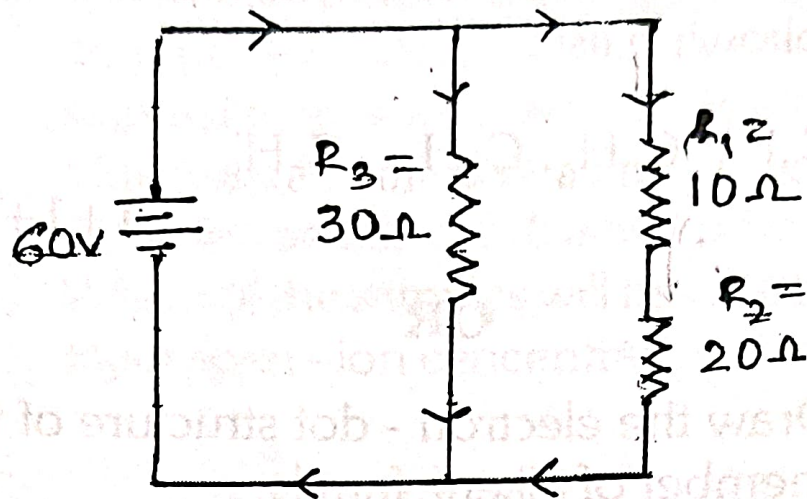
- i) Draw the electron - dot structure of the 1st member of alkene family.
- ii) "Carbon forms a large number of covalent compounds" — Mention two factors to justify the above statement.
- iii) How can you distinguish between a saturated and an unsaturated hydrocarbon by a chemical test? [1+1+1=3]

Section - C (Physics)

15. A convex lens with a focal length of 12 cm has object at a distance of 20 cm in front of the lens. A blurred image is obtained on the screen placed at a distance of 23 cm in front of the lens. In order to obtain a clear image, the screen has to be moved :

- a) towards the lens ~~b) away from the lens~~
- c) to a position very far from the lens
- d) either towards or away from the lens. [1]

16.



In the circuit given, $R_1 = 10\Omega$, $R_2 = 20\Omega$, $R_3 = 30\Omega$ find the correct statement between the flow of current through 10Ω , 20Ω and 30Ω resistors:

- a) Current through $R_1 >$ Current through 20Ω
- b) Current through $R_3 <$ Current through 20Ω
- ~~c) Current through $R_1 =$ Current through 20Ω~~
- d) Current through $R_3 >$ Current through 20Ω

[1]

The following question consists of two statements — Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below

- a) Both A & R are true and R is the correct explanation of A.
- ~~b) Both A & R are true and R is not the correct explanation of A.~~

- c) A is true but R is false.
- d) A is false but R is true.
17. Assertion (A) : The magnetic field lines around a current carrying straight wire do not ~~interest~~ ^{intersect} each other.
- Reason (R) : The ^{magnitude} ~~magnetic~~ of the magnetic field produced at a given point increases as the current through the wire increases. [1]
18. Why is tungsten used almost exclusively for filament of electric lamps? [2]

OR

Why are the conductors of electric heating devices, such as bread toaster and electric irons made of an alloy rather than a pure metal. [2]

19. How does the scattering of light of different wavelength affected by the size of particles. Why the colour of sky is blue. [3]

OR

The near point of a person suffering from hypermetropia is 75 cm. Calculate the focal length and power of the lens required to enable him to read the newspaper which is kept at a distance of 25 cm from the eye. State any one cause of this defect. ^{2. Tob} ^{→ 37.5 cm} [3]

20. i) Three resistors having values R_1 , R_2 and R_3 are connected in series to a battery. Suppose R_1 carries a current of 2.0A , R_2 has a resistance of 3.0Ω and R_3 dissipates 6.0 watts of power. What is the voltage across R_3 . 3V

ii) Two resistors A and B of resistance 4Ω and 6Ω respectively are connected in parallel. The combination is connected across a 6V battery. Calculate the power dissipated in each resistor. [5]

$$A \rightarrow 9\text{W}$$

$$B \rightarrow 1\text{W}$$

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$$\frac{6 \times 6}{6} = 6$$